

EINDHOVEN AP, Netherlands

WMO: 063700

Lat: 51.4497N

Lon: 5.3769E

Elev: 74

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.7	23.5	13.1	10.8	23.6	17.6	13.4	25.6	27.8	49.1	25.3	48.9	5.7	70	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.3	68.0	82.1	66.6	78.5	64.9	70.1	81.8	68.3	78.7	66.6	75.6	7.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.3	74.1	64.6	91.5	72.2	63.1	86.8	70.4	34.0	82.1	32.6	78.8	31.2	75.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	19.6	17.3	DB	13.3	93.0	6.1	3.2	8.9	95.3	5.3	97.1	1.9	98.9	-2.6	101.2	(4)
(5)			WB	12.7	73.7	6.1	1.7	8.3	74.9	4.8	76.0	1.3	76.9	-3.1	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	38.0	39.1	43.8	50.0	56.6	61.7	65.5	64.7	59.0	52.3	44.5	39.1	(6)	
	DBStd	11.79	7.75	7.26	6.41	6.85	6.59	5.94	5.99	5.36	5.11	6.20	6.44	7.58	(7)	
	HDD50	1568	376	308	209	83	14	0	0	0	1	48	185	344	(8)	
	HDD65	5224	838	726	658	450	273	134	67	71	193	395	615	804	(9)	
	CDD50	2024	2	3	16	84	218	351	482	455	270	118	20	5	(10)	
	CDD65	206	0	0	0	1	12	35	84	61	12	1	0	0	(11)	
	CDH74	2006	0	0	0	33	146	374	783	564	100	6	0	0	(12)	
	CDH80	578	0	0	0	2	24	102	251	181	18	0	0	0	(13)	
(14) Wind	WSAvg	8.2	9.7	9.3	9.0	8.0	7.8	7.5	7.2	6.8	7.1	7.9	8.2	9.3	(14)	
(15) Precipitation	PrecAvg	32.0	2.8	2.6	2.0	1.8	2.4	2.8	3.1	3.1	2.6	2.6	3.0	3.2	(15)	
	PrecMax	40.8	5.7	5.8	4.1	3.3	4.4	8.1	5.7	6.7	6.6	7.0	5.1	6.1	(16)	
	PrecMin	24.3	0.1	0.6	0.2	0.0	0.7	0.5	1.0	0.9	0.4	0.9	0.6	1.1	(17)	
	PrecStd	3.7	1.3	1.3	1.1	0.8	1.0	1.5	1.4	1.8	1.6	1.1	1.1	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	59.0	67.3	78.7	83.3	89.0	92.5	90.4	82.4	73.6	62.9	56.7	(19)	
		MCWB	52.2	50.0	54.6	59.0	65.1	69.5	69.2	70.0	66.6	63.2	57.3	53.4	(20)	
	2%	DB	53.0	54.2	61.7	72.3	77.9	82.5	86.5	85.1	76.7	67.8	58.6	54.5	(21)	
		MCWB	50.3	49.2	51.6	56.9	61.9	66.3	68.4	67.9	64.6	60.7	54.9	51.9	(22)	
	5%	DB	50.6	51.7	57.4	67.1	74.0	78.2	82.0	80.4	72.1	64.1	56.3	52.2	(23)	
		MCWB	48.1	48.0	50.1	54.8	60.1	64.5	66.9	66.7	62.9	59.0	53.3	49.9	(24)	
	10%	DB	48.2	49.4	53.9	62.4	69.6	74.1	78.0	75.7	68.3	61.3	53.8	49.7	(25)	
		MCWB	45.9	46.6	48.7	52.7	58.4	62.6	65.6	64.6	60.8	57.3	51.3	47.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	52.2	55.8	60.7	67.5	71.6	72.6	72.5	68.5	64.0	58.0	54.5	(27)	
		MCDB	54.7	55.4	63.4	72.3	78.7	83.8	84.9	85.2	78.4	70.9	60.9	55.8	(28)	
	2%	WB	50.5	50.3	53.3	58.0	64.2	68.2	70.2	69.7	66.0	61.6	55.7	52.3	(29)	
		MCDB	52.5	53.1	59.0	68.7	74.3	78.7	82.0	81.1	73.9	66.4	58.2	54.2	(30)	
	5%	WB	48.3	48.7	51.3	55.9	61.5	66.0	68.5	67.8	64.0	59.7	53.7	50.2	(31)	
		MCDB	50.4	51.4	55.9	65.4	70.1	75.1	79.0	77.3	70.2	63.4	55.9	52.0	(32)	
	10%	WB	46.2	46.6	49.3	53.8	59.4	63.9	66.6	65.9	62.1	57.9	51.4	47.8	(33)	
		MCDB	48.1	49.1	53.4	60.9	67.5	72.3	75.6	73.7	67.1	61.0	53.5	49.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.1	14.5	18.0	18.1	18.3	17.9	18.1	16.6	13.8	10.5	8.8	(35)	
		MCDBR	9.9	13.6	21.1	25.4	25.1	25.2	24.6	25.1	21.6	16.8	11.6	9.6	(36)	
	5% WB	MCWBR	8.7	9.7	12.7	12.7	11.8	11.7	10.2	10.8	11.1	10.6	8.8	8.3	(37)	
		MCWBR	9.6	11.6	17.5	21.8	21.1	21.8	21.6	21.9	18.9	15.4	10.9	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.360	0.393	0.423	0.431	0.427	0.435	0.425	0.398	0.373	0.349	0.333	(40)	
		taud	2.344	2.307	2.221	2.177	2.203	2.229	2.211	2.260	2.325	2.372	2.352	2.319	(41)	
	Ebn at Noon		210	238	251	258	262	264	259	254	248	230	203	189	(42)	
		Edh at Noon	21	28	36	42	43	42	43	38	32	26	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		238	453	823	1321	1587	1717	1610	1374	1016	596	278	182	(44)	
	RadStd		29	88	117	143	132	138	193	122	94	47	28	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page